

CASE STUDY

Registrar Document Request System Using VB.NET and MySQL

Development Period: 2 Months

Project Type: Group Project

Technology:

- VB.NET Windows Forms
 - MySQL Database
 - Visual Studio
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I. CASE STUDY

The Registrar's Office is responsible for maintaining student academic records and processing requests for official school documents. Students regularly request documents such as Transcript of Records, Certificate of Enrollment, Certificate of Good Moral, Certification, and other registrar-related documents.

At present, the Registrar's Office wants to improve the way document requests are recorded and monitored. Manual recording makes it difficult to track requests, identify pending documents, monitor payments, and determine which documents are already ready for release.

The school has therefore decided to develop a simple **Registrar Document Request System**.

Your group has been assigned as the development team responsible for creating the system using **VB.NET and MySQL**.

The system should provide a simple and organized way to manage students, document types, and document requests.

II. PROJECT OBJECTIVE

The main objective of this project is to develop a functional **Registrar Document Request System** using VB.NET and MySQL.

The system should demonstrate the group's ability to apply the programming and database concepts learned during the course.

Specifically, the system should be able to:

1. Manage student information.
2. Manage available registrar documents.
3. Create document requests.

4. Calculate the total amount of a request.
 5. Monitor the status of document requests.
 6. Record payment information.
 7. Search and view requests.
 8. Generate simple reports.
 9. Store information in a MySQL database.
 10. Implement a user login system.
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III. PROJECT GROUPING

This project will be completed using the **same groups assigned for the Final Examination**.

Students are not required to create new groups.

Each member of the group must have a specific responsibility and must contribute to the development of the system.

The instructor may ask any member of the group to explain any part of the system during the final presentation.

Therefore, **all members must understand how the complete system works**, not only the part assigned to them.

IV. SYSTEM USERS

To keep the project attainable within two months, the system will have only two main user roles.

Administrator

The Administrator can:

- Manage users.
- Manage students.
- Manage document types.
- View document requests.
- Generate reports.

Registrar Staff

The Registrar Staff can:

- Search students.

- Create document requests.
 - View requests.
 - Record payment information.
 - Update request status.
 - View reports.
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V. REQUIRED SYSTEM MODULES

The group is required to develop the following modules.

1. Login Module

The system must have a login form.

Required fields:

- Username
- Password
- Login button

The system should check the username and password from the MySQL database.

The system should also identify the user's role.

Example:

Administrator

or

Registrar Staff

2. Main Menu

After successfully logging in, the user should be redirected to the Main Menu.

The Main Menu should provide access to:

- Student Management
- Document Management
- Document Requests
- Reports
- User Management

- Logout

The system does not need a complicated dashboard.

A simple and organized menu is sufficient.

3. Student Management

The system should allow authorized users to manage student records.

Required information:

- Student ID
- LRN
- Last Name
- First Name
- Middle Name
- Course
- Year Level
- Section
- Contact Number

Required functions:

- Add
- Edit
- Search
- Delete or Deactivate
- View

The student records should be stored in MySQL.

4. Document Management

The system should allow the Registrar to manage the documents available for request.

Example:

Document	Fee
Transcript of Records	150.00
Certificate of Enrollment	50.00
Certificate of Good Moral	100.00
Certification	50.00
Honorable Dismissal	100.00

Required information:

- Document ID
- Document Name
- Description
- Fee
- Status

Required functions:

- Add
- Edit
- Search
- Activate/Deactivate

The document fee must come from the database.

5. Document Request Module

This is the **main module of the project**.

The Registrar Staff should be able to create a new document request.

The request form should contain:

Request Information

- Request Number
- Request Date
- Student ID
- Student Name

- Course
- Year Level
- Document
- Quantity
- Amount
- Total Amount
- Payment Status
- Request Status
- Processed By

The system should automatically generate the Request Number.

Example:

REQ-2026-00001

The student information should be retrieved from the database after searching for the Student ID.

The system should automatically calculate:

Subtotal = Document Fee × Quantity

and

Total Amount = Sum of all Subtotals

6. Request Status

The system should use the following basic statuses:

Pending

The request has been created but has not yet been processed.

Processing

The Registrar is currently preparing the requested document.

Ready for Release

The document is already available for the student to claim.

Released

The document has already been released to the student.

Cancelled

The request was cancelled.

These statuses should be updated by authorized users.

7. Payment Information

The system should allow the Registrar Staff to record basic payment information.

Required fields:

- Payment Status
- OR Number
- OR Date
- Amount Paid

Payment status may be:

- Unpaid
- Paid

Students do not need to develop an online payment system.

The project only requires recording payment information.

8. Request List

The system should display document requests using a **DataGridView**.

Suggested columns:

Request No.	Date	Student ID	Student Name	Document	Amount	Status
REQ-2026-00001	08/18/2026	20260001	Juan Dela Cruz	TOR	150.00	Pending
REQ-2026-00002	08/18/2026	20260002	Maria Santos	Good Moral	100.00	Processing

The user should be able to:

- Search
 - Select a request
 - View details
 - Update status
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9. Reports

The system only needs **simple reports**.

The group must provide at least three of the following:

1. All Document Requests
2. Pending Requests
3. Released Requests
4. Requests by Date
5. Requests by Document Type
6. Payment Report

The report may be displayed in a DataGridView or another appropriate VB.NET reporting format.

Students are **not required to develop an advanced reporting system**.

VI. REQUIRED DATABASE

Create a MySQL database named:

registrar_db

The minimum required tables are:

tblusers

- UserID
- Username
- Password
- FullName
- Role
- Status

tblstudents

- StudentID
- LRN
- LastName
- FirstName
- MiddleName

- Course
- YearLevel
- Section
- ContactNo
- Status

tbldocuments

- DocumentID
- DocumentName
- Description
- Fee
- Status

tblrequest

- RequestID
- RequestNo
- StudentID
- RequestDate
- TotalAmount
- PaymentStatus
- ORNo
- ORDate
- Status
- CreatedBy

tblrequestdetails

- RequestDetailID
- RequestID
- DocumentID
- Quantity
- Amount
- SubTotal

The group may add additional tables if necessary, but these five tables are the **minimum required tables**.

VII. DATABASE RELATIONSHIPS

The database should demonstrate proper relational database design.

Basic relationships:

Student

↓

Request

↓

Request Details

↓

Document

For example:

One student can have many requests.

One request can contain one or more documents.

One document can appear in many different requests.

The group must create an **Entity Relationship Diagram (ERD)** showing these relationships.

VIII. MINIMUM PROJECT REQUIREMENTS

To keep the project achievable within two months, the following are the minimum requirements.

Required

- Login
- MySQL database
- Database connection
- Student CRUD
- Document CRUD
- Create document request
- Automatic request number

- Automatic amount computation
 - Request list
 - Search function
 - Request status update
 - Payment information
 - At least 3 reports
 - DataGridView
 - Input validation
 - Parameterized SQL queries
 - ERD
 - Sample data
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IX. MINIMUM SAMPLE DATA

Before the final presentation, the system should contain:

- At least **15 students**
- At least **5 document types**
- At least **3 user accounts**
- At least **15 document requests**

The requests should demonstrate different statuses.

Example:

- 3 Pending
 - 3 Processing
 - 3 Ready for Release
 - 3 Released
 - 3 Cancelled
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X. TWO-MONTH DEVELOPMENT PLAN

The project will be developed gradually.

Weeks 1–2: Planning and Database

The group should:

- Understand the case study.
- Identify system requirements.
- Design the system flow.
- Design the database.
- Create the ERD.
- Create MySQL tables.
- Insert initial sample data.

Expected output:

Database + ERD + system flow

Weeks 3–4: VB.NET Foundation

The group should:

- Create the Visual Studio project.
- Create the Login Form.
- Connect VB.NET to MySQL.
- Create the Main Menu.
- Create Student Management.
- Create Document Management.

Expected output:

Working Login + Student + Document modules

Weeks 5–6: Document Request

The group should focus on the main transaction.

Develop:

- New Request
- Student Search
- Document Selection
- Quantity

- Automatic computation
- Request Number
- Saving the request
- Request List
- Request Details

Expected output:

Working Document Request module

Week 7: Payment, Status, and Reports

Develop:

- Payment information
- OR Number
- OR Date
- Request Status
- Search/filter
- At least three reports

Expected output:

Complete core system

Week 8: Testing and Finalization

The group should:

- Test every function.
- Fix errors.
- Check database relationships.
- Improve the interface.
- Add validation.
- Prepare documentation.
- Prepare presentation.
- Prepare final demonstration.

Expected output:

Completed and tested system

XI. FINAL SYSTEM DEMONSTRATION

During the final presentation, the group should demonstrate the following scenario:

Step 1

Login using an Administrator account.

Step 2

Show the Main Menu.

Step 3

Show the Student Management module.

Step 4

Search for a student.

Step 5

Create a new document request.

Step 6

Select at least two documents.

Step 7

Show the automatic calculation of the total amount.

Step 8

Save the request.

Step 9

Show the generated Request Number.

Step 10

Open the request from the request list.

Step 11

Enter the OR Number and OR Date.

Step 12

Change the Payment Status to Paid.

Step 13

Change the Request Status to Processing.

Step 14

Change the status to Ready for Release.

Step 15

Change the status to Released.

Step 16

Search for the released request.

Step 17

Show at least three reports.

Step 18

Explain the database tables and relationships.

XII. PROJECT LIMITATIONS

To ensure that the project can be completed within two months, the following are **not required**:

- Online student portal
- Online payment
- SMS notification
- Email notification
- Mobile application
- Cloud database
- Artificial intelligence
- QR code
- Biometric system
- Advanced accounting system
- Advanced inventory system
- Online document upload
- Complex dashboard
- Multi-branch support

Groups may add additional features if the required features are already completed.

The priority is to produce a **working and reliable system**, not a system with many unfinished features.

XIII. GROUP MEMBER RESPONSIBILITIES

Each group should assign responsibilities.

Suggested roles:

Project Leader

Responsible for:

- Coordinating the group
- Monitoring progress
- Integrating the modules
- Preparing the final presentation

Database Developer

Responsible for:

- MySQL database
- Tables
- Relationships
- SQL queries
- ERD

VB.NET Developer

Responsible for:

- Forms
- Database connection
- CRUD operations
- Transactions

UI Developer

Responsible for:

- Form layout

- Navigation
- User interface consistency

Tester/Documenter

Responsible for:

- Testing
- Error checking
- Documentation
- Screenshots
- User manual

A group member may have more than one responsibility depending on the number of members.

XIV. IMPORTANT RULE

Although responsibilities may be divided among group members, **every member must understand the complete system.**

During the final presentation, the instructor may ask any member to:

- Explain the database.
- Explain an SQL query.
- Explain the VB.NET code.
- Demonstrate a module.
- Explain the relationships.
- Fix or modify a part of the system.

A member who only knows his/her assigned module may have difficulty during the final evaluation.

XV. DELIVERABLES

Each group must submit:

1. VB.NET source code
2. MySQL SQL file
3. ERD
4. System documentation

5. User manual
6. Screenshots of the system
7. Final presentation
8. Working system for demonstration

The SQL file must be able to recreate the database and tables.

XVI. PROJECT GRADING

Criteria	Percentage
Database Design and ERD	20%
VB.NET/MySQL Functionality	25%
Document Request Transaction	20%
CRUD and Search Functions	10%
Validation and Error Handling	10%
User Interface	5%
Final Demonstration	10%
TOTAL	100%

XVII. FINAL PROJECT GOAL

The purpose of this project is not simply to create a Windows application.

The project should demonstrate that the group can apply the concepts learned throughout the course to develop a **realistic database-driven information system**.

By completing the project, students should demonstrate their ability to:

Analyze a problem → Design a database → Create a VB.NET application → Connect to MySQL → Perform CRUD operations → Process transactions → Generate information → Present a working system.

The completed project should provide the Registrar's Office with a simple, organized, and functional way of managing student document requests.